Community Software Project - OpenFOAM

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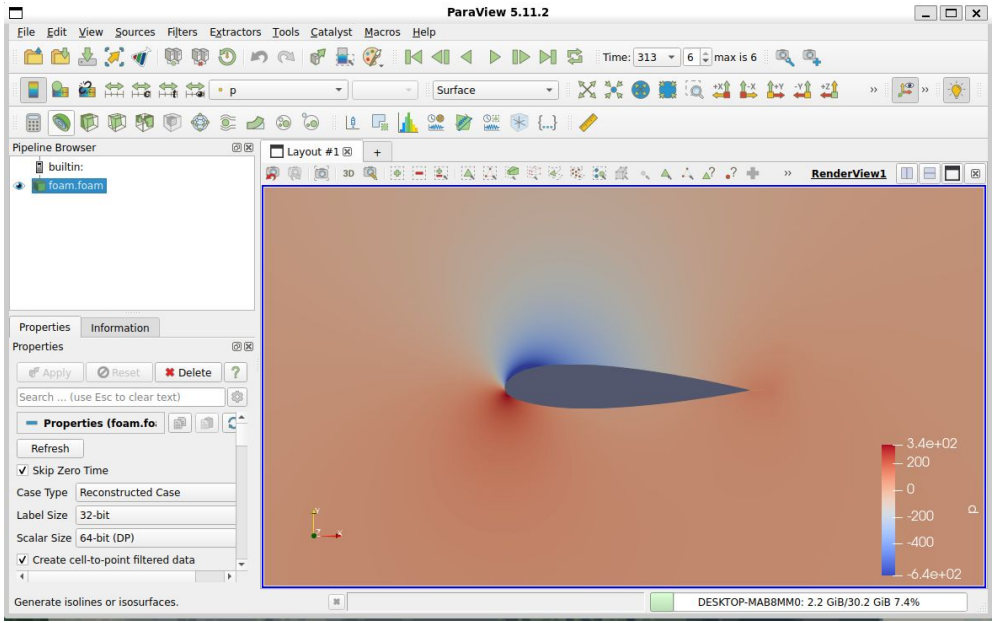
What is OpenFOAM?

- “Open source Field Operation And Manipulation”
- Open source solution for computational fluid dynamics written in C++
- Prominent throughout industry engineering and research
- Can be used for 2D/3D CFD calculations and optimization
- Licensed under GPL v3

Example

```
tutorials > incompressibleFluid > airFoil2D > log.foamRun
/*---------------------------------------------------------------------------*\
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Version:  dev
     \\/     M anipulation  |
\*---------------------------------------------------------------------------*/
Build  : dev-e2c120669f96
Exec   : foamRun
Date   : Mar 30 2026
Time   : 18:24:39
Host   : "DESKTOP-MAB8MM0"
PID    : 66710
I/O    : uncollated
Case   : /home/april/OpenFOAM/OpenFOAM-dev/tutorials/incompressibleFluid/airFoil2D
nProcs : 1
sigFpe : Enabling floating point exception trapping (FOAM_SIGFPE).
fileModificationChecking : Monitoring run-time modified files using timeStampMaster (fileModificationSkew 10)
allowSystemOperations : Allowing user-supplied system call operations

// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //
Create time

Create mesh for time = 0

Selecting solver incompressibleFluid
Selecting viscosity model constant
Selecting turbulence model type RAS
    Selecting RAS turbulence model SpalartAllmaras
No MRF models present


SIMPLE: Convergence criteria found
        p: tolerance 1e-05
        U: tolerance 1e-05
        nuTilda: tolerance 1e-05


SIMPLE: Operating solver in steady-state mode with 1 outer corrector
SIMPLE: Operating solver in SIMPLE mode
```




```
Time = 313s

smoothSolver:  Solving for Ux, Initial residual = 6.75419e-06, Final residual = 6.41637e-07, No Iterations 2
smoothSolver:  Solving for Uy, Initial residual = 3.0668e-06, Final residual = 1.32396e-07, No Iterations 4
GAMG:  Solving for p, Initial residual = 7.38e-06, Final residual = 7.84296e-07, No Iterations 4
time step continuity errors : sum local = 4.89079e-09, global = -1.1675e-09, cumulative = 0.00025185
smoothSolver:  Solving for nuTilda, Initial residual = 9.86613e-06, Final residual = 9.79795e-07, No Iterations 2
ExecutionTime = 2.11471 s  ClockTime = 2 s


SIMPLE solution converged in 313 iterations

End
```

Who Manages OpenFOAM?

CFD Direct
The Architects of OpenFOAM

- “CFD Direct is the manager and primary developer of OpenFOAM, on behalf of the OpenFOAM Foundation, the copyright holder and licensor of OpenFOAM.”
- Also sees support from community members
- There exist OpenFOAM “spinoffs” forked and developed by third party community members

Who uses OpenFOAM?

- Utilized in industry for engineering applications as well as research
- Seemingly very common in automotive development in Europe
- Relied on and directly supported by universities and other private companies
- OpenFOAM conferences happen once a year also supported by those who utilize the software

Prominent Contributors

- Henry Weller - Co-founder of OpenFOAM, core architect/developer
- Chris Greenshields - Co-founder of OpenFOAM, managing director
- Will Brainbridge - Full-time core developer
- Timo Niemi & Juho Peltola - Frequent contributors from VTT
- Other organizations have signed contribution agreements and have development history

Development Activity

- Code hosted primarily on GitLab
- Sees frequent updates and messages
- Commits <1000 lines

Masaru Sato @satorccm — 1 hour ago
Commented on issue #3508 "the sign for eqn.diag() and eqn.source() for coded source term is reversed" at OpenFOAM / Core / OpenFOAM
Hi, Kuti...

Kutalmış Berçin @KutalmisBercin — 10 hours ago
Pushed to branch `main` at OpenFOAM / Community / TC - Turbulence / turbulence-community
`e78c7232` · SUBMODULE: update EllipticBlending, HelicitySA, gammaSST, kOmegaSSTPDA

Kutalmış Berçin @KutalmisBercin — 11 hours ago
Pushed to branch `develop` at OpenFOAM / Core / OpenFOAM
`9219f3d2` · Merge branch 'tutorial_Allclean_fix' into 'develop'
... and 1 more commit. Compare `549b6d29...9219f3d2`

Kutalmış Berçin @KutalmisBercin — 11 hours ago
Accepted merge request !805 "tutorials: source CleanFunctions in Allclean for sloshingTank2D in interFoam" at OpenFOAM / Core / OpenFOAM

Kutalmış Berçin @KutalmisBercin — 11 hours ago
Approved merge request !805 "tutorials: source CleanFunctions in Allclean for sloshingTank2D in interFoam" at OpenFOAM / Core / OpenFOAM

Kutalmış Berçin @KutalmisBercin — 11 hours ago
Commented on merge request !805 "tutorials: source CleanFunctions in Allclean for sloshingTank2D in interFoam" at OpenFOAM / Core / OpenFOAM
@OpenFoe - Hi David, many thanks for fixing the bug; much appreciated it.

Robert Perry @skoloCFD — 14 hours ago
Commented on merge request !797 "Draft: yPlus field entry to allow user-selection of U field used" at OpenFOAM / Core / OpenFOAM
Done in latest commit....

Kutalmış Berçin @KutalmisBercin — 15 hours ago
Commented on issue #3515 "snappyHexMesh option to only write final mesh, no intermediate steps" at OpenFOAM / Core / OpenFOAM
@ext-mjanssens - would you mind comment on it?

Contributing

- Requires legal agreement from both parties
- Gives short advice on how to contribute
- Includes both code quality and formatting guidelines
- Have their own site for bug/issue reporting

Contribution Idea

- Some documentation looks incomplete in some examples with blank fields

```
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Application
    electrostaticFoam

Description
    Solver for electrostatics.

\*---------------------------------------------------------------------------*/
```

```
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\*---------------------------------------------------------------------------*/
```

Questions?